

Nicholas H. David

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CAREER OBJECTIVE

Graduate Biomedical Engineer with a sub-major in Mechanical Engineering and hands-on experience across medical devices, biomaterials, microfabrication, CAD, electronics and data analysis. Experienced in laboratory, cleanroom and project-based environments, with a strong focus on practical problem-solving, technical troubleshooting and multidisciplinary engineering. Skilled in translating engineering concepts into functional prototypes, experimental workflows and data-driven solutions, with a strong interest in medical technology, product development and healthcare innovation.

EDUCATION

UNIVERSITY OF TECHNOLOGY SYDNEY

Bachelor of Engineering (Honours)

Major in Biomedical Engineering; Sub-Major in Mechanical Engineering

Cumulative GPA: 6.48/7.0

Dean's List 2025 & 2026

Sydney, NSW

Feb 2023 - Oct 2026

PROFESSIONAL EXPERIENCE

INSTITUTE OF BIOMEDICAL MATERIALS AND DEVICES

Research Fellow

Ultimo, NSW

Jun 2026 - Present

- Conduct multidisciplinary engineering research involving biomedical device development, photolithography, microfabrication and experimental testing.
- Analyse experimental results and troubleshoot technical and manufacturing challenges to support evidence-based design improvements.
- Collaborate with researchers across engineering and scientific disciplines and communicate technical findings and project outcomes.
- Prototype components through 3D printing and microfabrication, evaluating fit, function and manufacturability before refining designs based on experimental testing.

MYER

Distribution Centre Team Leader

Eastern Creek, NSW

Feb 2023 - Jun 2026

- Developed automation tools that streamlined repetitive inventory processes, reducing daily processing time by approximately 2-3 hours.
- Led and coordinated a team of 10 employees, allocating resources and priorities to maintain operational performance.
- Developed process improvements and workflow optimisation strategies, demonstrating systems-thinking applicable to engineering environments.
- Trained and onboarded 25+ employees across operational systems and procedures.
- Staff Australia Myer Employee of the Year Award 2024.

DAVIDS MECHANICAL REPAIRS

Junior Mechanic

Smithfield, NSW

Dec 2018 - Feb 2023

- Built effective communication skills by clearly explaining complex mechanical issues to non-technical clients.
- Collaborated with qualified mechanics to troubleshoot and resolve complex mechanical faults, developing team-based problem solving skills.

SELECTED PROJECTS

SEIZURE PREDICTION MEDICAL DEVICE

- Led a multidisciplinary team in developing a concept for a seizure prediction device, coordinating design decisions and project responsibilities.
- Evaluated regulatory requirements, including TGA and FDA considerations, and contributed to verification and validation planning.

PULSE OXIMETER - ARDUINO / MAX30102

- Built and debugged a pulse oximeter prototype using an Arduino UNO and MAX30102 sensor to display heart rate and SpO₂; troubleshoot sensor communication, signal-processing code and LCD/serial output, calibration

3D-PRINTED SCAFFOLD FOR OSTEOSARCOMA RECONSTRUCTION

- Collaborated to design and evaluate a 3D-printed orthopaedic scaffold for distal femur reconstruction following osteosarcoma resection.
- Applied computational modelling, material analysis and physical testing to compare design options and support evidence-based decisions.

NANOPARTICLE-BASED CANCER CELL DETECTION

- Designed and conducted experiments investigating nanoparticle-based cancer cell detection and analysed experimental results using ImageJ.
- Interpreted quantitative imaging data to identify differences between control and test samples and communicate findings.

UTS TECH FESTIVAL 2025 & 2026 - BIOMEDICAL ENGINEERING SHOWCASE

- Selected to present a biomedical engineering project combining 3D bioprinting, cell culture and fluorescence microscopy to model chronic traumatic encephalopathy.
- Recognised as an outstanding student project at the UTS Tech Festival.

SKILLS

Data & Analysis: Python, Java, Microsoft Excel, MATLAB, data analysis, experimental data interpretation, ImageJ

Engineering & Technology: SolidWorks, Fusion 360, Ansys, 3D modelling, additive manufacturing, LabVIEW

Project & Professional: Process improvement, project coordination, stakeholder communication, technical reporting, risk assessment, full driver's license and access to vehicle

Interpersonal: Leadership, teamwork, problem-solving, communication, time management

REFERENCES

Available upon request.